

SMART AI BASED ATTENDANCE MONITORING SYSTEM USING YOLOV8

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Abstract—The proposed system utilizes the YOLOv8 – a computer vision model for object detection, classification and instant segmentation of captured images. The proposed work helps to identify and extract facial regions from the input images or real-time videos through the fixed CCTV camera in the classrooms and the mark the attendance for the students. The current attendance system makes it difficult to track the student attendance during each lecture period and it also consumes lot of time. There is a chance for human errors also to occur. The proposed system uses YOLOv8 algorithm, that compares the registered faces from the database and identifies the individual identities. The algorithm employs a CNN that divides the captured image into grid. Hence, the proposed system has a friendly user interface, that enables the administrators to manage the record of student's attendance easily and helps to save time and reduce human errors. It makes attendance monitoring more easier and faster. The used algorithm performs faster than Faster R-CNN in terms of speed and accuracy. Its GPU latency compared to R-CNN is 1.3ms. The rate of skipping classes and students leaving early can be reduced greatly using the proposed work. Experimental data shows that the accuracy rate of video-monitoring system is up to 80%. Hence the proposed system will help in the reduction of errors and compute the attendance data faster compared to any other real-time monitoring algorithms.

Keywords— Face detection, YOLOv8, Deep Learning, AI, Attendance Monitoring. Real Time system

I. INTRODUCTION

Whether in class or at a work meeting, the attendance process takes up a considerable amount of time from the teaching or conference activity itself, especially when there are large numbers of people. In modern society, people need advanced solutions for this time-consuming task. Activities that anyone can do anytime, anywhere [1]. During the covid-19 pandemic the education system has shifted from online classes to online classes, but keeping task of attendance in offline as well as online was a time-consuming task. To overcome the above issue, many systems like biometric attendance system, QR codes were introduced. But the early system was not much efficient and did not track the attendance accurately [2]. Face recognition attendance system is an accurate and helps to identify the individual in a fraction of second. One of the cost-effective technique is using Haar cascade. It is a user-friendly machine learning

based algorithm. It combines the Haar cascade along with OpenCV2. It accurately matches the images and finds the output using advanced edge computing techniques [3]. In many universities attendance is still done by just passing an attendance sheet to the students and signed by the students. They either call students name or pass the attendance sheets. The system of interest detects each of the student entering into the classrooms at the doorstep. Exact position of the face is maintained to detect the students face and update in the database [4]. AI is a widely used technology these days. For the purpose of managing attendance, a dependable database is employed, together with an algorithm called FaceNet that was created by Google and the MQTT protocol for sophisticated and secure data transfer. The method for taking attendance involves attaching the system at a 45-degree angle to the door hinges at the classroom entry [5]. Because of its numerous practical uses, face recognition technology is significant in a variety of fields. This paper presents a novel face recognition system that smoothly combines support vectors, VGG Face for feature extraction, and multi-task cascaded convolutional neural network (MTCNN) for precise face detection [6]. The system must achieve accurate and timely performance in reviewing work management for each mechanism. It is based on human approval and uses depth calculation methods. To execute personnel inspection, pattern inspection, guarantee the standards and availability of personnel recognition, and increase system efficiency, processing, information dilib, cvzone, and YOLO calculation methods are coupled [7]. Python is used in conjunction with object-oriented analysis and design technique to detect faces in live streaming recordings of the classroom [8]. My SIMS is intended to be a portable teaching tool. It has offered the school practice approach a whole new solution. There is a way for figuring out whether or not a person can be recognised for the choosing of products. Each student has a unique and different structure, so there is no lawless system of deception [9]. Even now, management review work is a crucial and demanding job. Technology has advanced to the point where storage technologies like RFID (Radiated Voice Identification) and biometric fingerprint drawing are available, but choosing which of these technologies to use will also require you to dedicate some time to thinking about it while at work. This suggested model computes the actual properties of humans and employs LBPH (Local Binary Cuboid Diagram) computation to distinguish between

distinct individuals. It also uses Haar series separators to identify different persons. possesses verification [10].

The algorithm is used in the suggested work. YOLOV8 is an automated, effective system that makes use of cutting-edge AI algorithms. It tracks attendance properly and effectively. Conventional techniques employed by universities and educational institutions are laborious and prone to human error. The technique makes use of an object detection method called YOLOV8 (You Only Look Once) to address the aforementioned problems. It has the capacity to quickly and accurately detect several things at once. The technology can identify faces in every lighting condition and angle by combining CNN and deep learning with YOLO. The goal of the proposed effort is to offer a powerful application that uses AI to improve the attendance system.

The paper's primary contribution is summarized here:

- The system automatically detects people and records their attendance by comparing the detected person with a pre-stored database.
- The renowned YOLOv8 algorithm is used to detect people in videos or images in real-time and with high accuracy.
- The architecture of YOLOv8 allows for fast and efficient processing even in hazardous environments, such as changing light and angles.
- The goal is to automate attendance monitoring using artificial intelligence, saving time and increasing convenience.

The following outlines the paper: Section 2 covers the literature review. Section 3 proposed the recommended methodology. Section 4 presents the discussion and outcomes. Section 5 summarizes the study's findings

II. RELATED WORKS

Olufemi S. Ojo et al. (2023) suggested due to its advanced quality, the published CNN-GA technology will enable a more standardized and more secure consideration system with better information on any storage system. This study shows that by developing an improved review system, an advanced security control system for knowledge provision can be achieved. Large amounts of true and small amounts of true yield higher accuracy [11].

The creation of an automated consideration system that improves communication consideration through surface inspection and approval approaches was detailed by Phul Babu Jha et al. (2023). The system, known as the LBPH system. The system's goal is to do rid of manual labour's inefficiencies and issues, like assigning tasks and wasting time [12].

A contactless attendance system that would be especially helpful during the COVID-19 pandemic was proposed by Deepti Saraswat et al. in 2023. Anti-spoofing techniques are employed by the system to stop erroneous attendance recording. It makes sure that attendance is being recorded by actual people and not by images or videos by determining how engaged a person is. This camera-based system is affordable, scalable, and appropriate for a range of establishments, including businesses and schools. Anti-

spoofing technology determines if a face is real or has been Photoshopped (using, for example, a picture). The solution makes use of Google's Firebase for storage, real-time processing, and the Google Vision API for face detection and recognition [13].

In response to the COVID-19 epidemic, Muhammad Haikal Mohd Kamil et al. (2023) created an online time and attendance system that makes use of mask detection and facial recognition technology. It trained the model using two types of masked face datasets: a conventional masked face dataset and a synthetic masked face dataset. It employed OpenCV for image processing and support vector machine (SVM) for face recognition. Mask detection, recognition, and model training are handled via Python programs. Users can upload selfies to the web-based system for processing in real time. Attendance management is contactless and effective thanks to face recognition accuracy of 81.8% and mask detection accuracy of 80% [14].

An intelligent search system based on the MTCNN (multi-managerial control system) handles personnel checks by utilising MTCNN power from Arc Face, according to Thai-Viet Dang et al. (2023). Arc Face employs depth studies to raise uniformity and enhance the system's inspection surface capabilities, especially in situations where challenging surface angles and lighting may exist. Additionally, this need is given to guarantee safety and scalability as well as to take job management into account [15].

In order to evaluate students' attentiveness levels in their execution, Zouheir Trabelsi et al. (2023) used the YOLOv5 algorithm to perform real-time execution and case recognition. The average standard degree, call rate, mAP and F1 fractions, and execution were all 76%. YOLOv5 was chosen with consideration for the balance between speed and standard degree, enabling the system to detect the presence or absence of the palate. It offers a quick comprehension of students' involvement [16].

Fisher Discriminant Analysis (LDA) was used by Hao Yang et al. (2020) to extract and separate features using Gabor features, perform feature expression using Gabor waves, and implement a human authentication system based on real-time video processing. By doing this, matching's robustness is increased. The Fisher approach, partially represented in grey, decreases calculation time and improves the authentication standard rate due to maximized between-group separability and minimal within-group variance. This system's standard rate is 82%, which both significantly increases and decreases work efficiency [17].

A system where faculty members use their smartphones to track attendance was suggested by Sikandar Khan et al. (2020). The region-based convolution network (RCNN), faster RCNN, and single-shot detector methods were employed. YOLO V3 algorithm was also used in the work. The classroom has a camera installed, and it takes pictures at the beginning and end of each lesson. Email notifications are used to transfer the data to the spreadsheets [18].

Using OpenCV, T. Patnaik et al. (2020) developed an attendance monitoring system. Convolutional neural networks (CNNs) with Haar Cascade classifiers were employed in the proposed work. The technology produced an

accurate and efficient output that removed human error and gave rise to a reliable attendance management system [19].

In 2020, Serign Modou Bah et al. proposed a new method that uses Local Binary pattern (LBH) along with Contrast adjustment, Bilateral filter, image blending, and Histogram equalisation. The authors took into consideration the varying light conditions in the classrooms, quality of the images captured, changes in the facial expressions and pose of the students. The project put into practice an actual automated attendance system [20].

III. PROPOSED METHODOLOGY

The suggested method is depicted in Figure 1, with each block described thoroughly beneath.

3.1. Input dataset

A range of face photos taken in a classroom setting make up the dataset utilised for AI attendance tracking. These pictures are essential for identifying and locating the faces of the pupils who are in the classroom. To ensure robustness, the dataset contains photos taken from various angles, with varying lighting conditions, and with varied student positions. For each student in this dataset, labelled data is also included to help with identification during real-time monitoring.

3.2. Pre-processing

Given the many external factors that affect the process of acquiring images, including lighting, background, and facial orientation, it is usually not uniform. To ensure that the image is appropriate for the YOLOv8 algorithm, the pre-processing step concentrates on improving the image quality. The photos must be converted to greyscale, noise must be removed, and contrast must be enhanced. Any noise is eliminated using Gaussian and median filters, and any unnecessary objects are taken out of the classroom environment using background subtraction techniques.

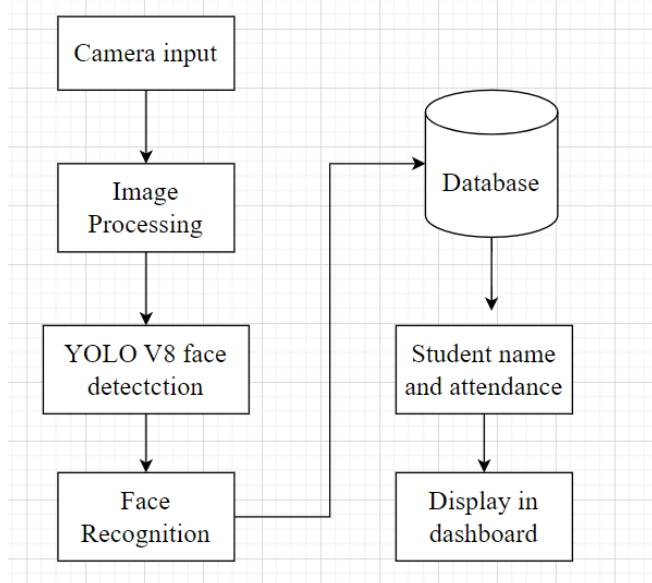


Fig. 1. Proposed method's block diagram

The figure 1 depicts the AI attendance monitoring system's methodology utilising YOLOv8. The input is first obtained from a camera and pre-processed to improve the quality of the image. After faces are detected by the YOLOv8 model, each student's identity is determined by a facial recognition method. The student's name and attendance status are updated, and attendance data is stored in a database. Lastly, a dashboard is used to conveniently display the attendance data.

3.3. Segmentation

YOLOv8 depends on precise picture segmentation to separate the region of interest (student faces) for effective face detection. Better bounding box detection is incorporated into the algorithm to improve face localisation. Adaptive thresholding is used in YOLOv8's segmentation process to distinguish faces from background. This is especially helpful in settings where there can be a number of pupils, different lighting, and obstacles like furniture or books.

3.4. Feature Extraction

The identification of the distinct features of every student's face depends heavily on feature extraction. Using deep learning-based feature extraction, the YOLOv8 system recognises important facial features such the mouth, nose, eyes, and general structure. These traits are then utilised to match the detected faces with the student records in the system's database. To increase recognition accuracy, the system collects additional characteristics from the camera, such as head direction and distance, in addition to facial landmarks. Additionally, YOLOv8 makes use of multi-scale feature maps and spatial pyramids to extract information from various image levels for reliable detection.

3.5. Classification

The method uses a neural network classifier that has been trained on the dataset to classify the observed faces after features have been extracted. A deep convolutional neural network (DCNN) with multiple layers is used by the YOLOv8 classification model for convolution, pooling, and fully connected classification. Through several training cycles, the model learns subtle distinctions across kids with similar face traits, enabling it to distinguish between them. Based on their attributes, the fully connected layer predicts each identified student's identification and compares it to the appropriate student ID stored in the system.

The suggested solution incorporates a specially trained model based on the YOLOv8 architecture, optimised for classroom attendance circumstances, to improve the classification process. This approach reduces the amount of misclassification brought on by occlusions, dim lighting, and overlapping faces. Additionally, YOLOv8 has been fine-tuned for real-time face detection, guaranteeing that each student's attendance is recorded accurately and on time throughout the class.

IV. RESULT AND CONCLUSION

The trained datasets contain registered students faces. The YOLOV8 algorithm efficiently classifies and identifies the exact student match from the database. The accuracy, sensitivity, precision, specificity and F1-score for the used algorithm is calculated using the below metrics. It is found that comparing to other algorithms the YOLOV8 algorithm has high accuracy and also more efficient.

(i) Accuracy:

This is a common metric for evaluating the effectiveness of classification models. The following formula is used to calculate accuracy, which is the ratio of properly identified samples to the total number of samples in the data set.

$$Accuracy = \frac{T_p + T_n}{T_p + T_n + F_p + F_n} \quad (1)$$

(ii) Precision: Precision is defined as the proportion of all classes that are both positively anticipated and positively experienced.

$$Precision = \frac{T_p}{T_p + F_p} \quad (2)$$

(iii) Sensitivity: This is the ability of the procedure to accurately identify the medical ailment or situation.

$$Sensitivity = \frac{T_p}{T_p + F_n} \quad (3)$$

(iv) Specificity: Specificity is referred to as True Negative Rate, it represents the proportion of all negative classes that the model classifier properly categorised as negative.

$$Specificity = \frac{T_n}{T_n + F_p} \quad (4)$$

(v) F1-score: The F1-score is a frequently employed metric in classification problems that offers a compromise between sensitivity and precision.

$$F1 \text{ score} = 2 \cdot \frac{Sensitivity \cdot Precision}{Sensitivity + Precision} \quad (5)$$

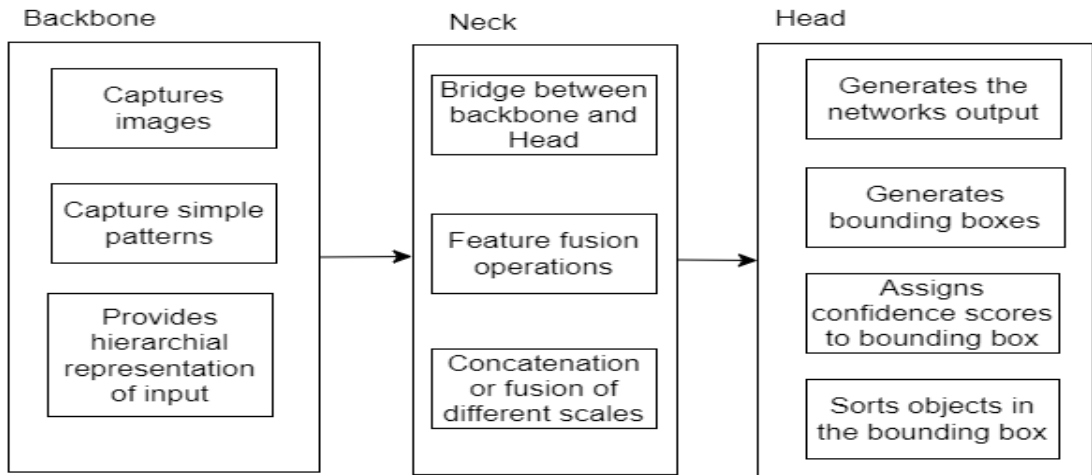


Fig. 2. Architecture of YOLOV8

The above figure 2 represents the YOLOV8 architecture used in the work. It consists of three main parts such as backbone, neck and head. The neck plays a major role in connecting both the backbone and head. The backbone is responsible for capturing images and simple patterns. It provides a hierarchical representation of the input. The neck does fusion operations. Finally the head generates the output and categorize according to the bounding box.

Table I. Student face detection and classification Efficiency

Algorithm	Accuracy	Specificity	F1-score	Precision	Sensitivity
RCNN	85.6	89.2	87	91.27	90.8
Haar cascade	93.4	91.5	92	92.5	93
YOLOV5	95.4	93.6	96.5	93.8	95.3
Proposed	97.5	98	98.5	97.8	98.7

The above Table 1 compares the accuracy, specificity, F1-score, Precision and Sensitivity parameters with different algorithm such as RCNN (Convolutional Neutral network), Haar cascade Algorithm, YOLOV5 algorithm and the work proposed YOLOV8 algorithm.

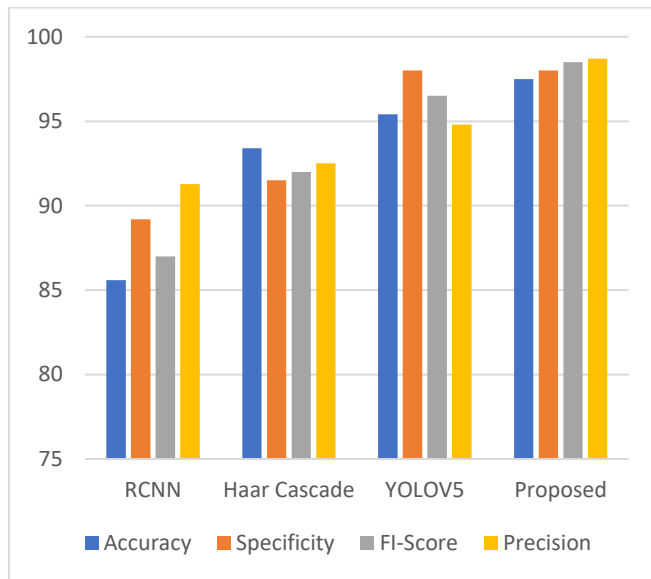


Fig. 3. Classification - performance comparison

The above figure 3 compares the different algorithms based on the accuracy, specificity and F1-score. The comparisons are in percentage.

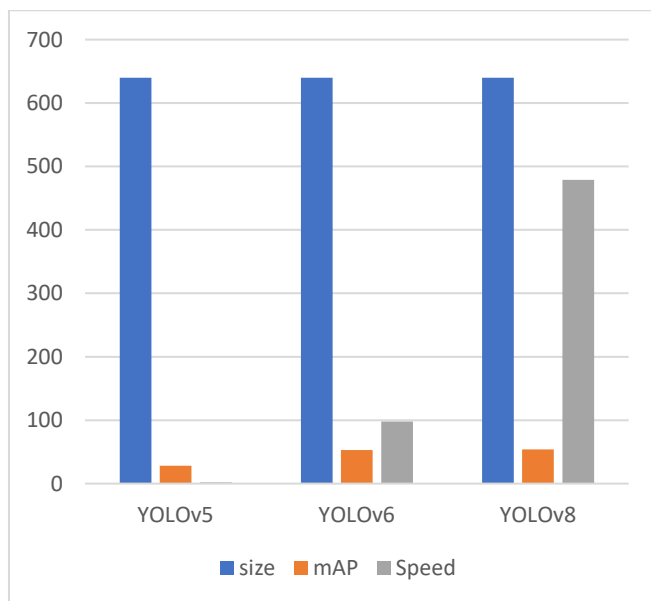


Fig. 4. Comparison of YOLO versions

The above figure 4 compares the different existing YOLO versions for object detection algorithms size(pixels), mAP, speed(ms) with the YOLOV8 algorithm.

V. CONCLUSION

The best way to compare face recognition systems is by comparing the advantages the YOLOv8 system offers in contrast to traditional methods. Although older approaches like Haar Cascade classifiers and CNNs ensure precise outcomes, the YOLOv8 method outsmarts the rest in performance and precision in real-time along with occlusions, varying lighting, and multiple faces in a frame. Detection in YOLOv8 is faster and more adaptive, providing less manual tuning based on deep learning. It also improves the robustness in larger settings since traditional algorithms could fail in detection speed or accuracy. In general, YOLOv8 provides an efficient, scalable, and reliable solution. Therefore, this system is quite suitable in applications where the environment has to be analyzed rapidly. Since it integrates harmoniously with applications having condition variations, it performs very well in real-world environments. It also successfully detects faces of multiple people at the same time without losing precision, and it is highly superior over other options to optimize school, university or workplace attendance management. The future work is to determine the student activity during the class hours, identifying the student mental state by analyzing the facial expressions. The AI based attendance monitoring is going to be very useful and helps consume time in the future.

REFERENCES

- [1] Tarmissi, K., Allaaboun, H., Abouellil, O., Alharbi, S., & Soqati, M. (2024, January). Automated Attendance Taking System using Face Recognition. In *2024 21st Learning and Technology Conference (L&T)* (pp. 19-24). IEEE
- [2] Shukla, A. K., Shukla, A., & Singh, R. (2024). Automatic attendance system based on CNN-LSTM and face recognition. *International Journal of Information Technology*, 16(3), 1293-1301.
- [3] Nguyen-Tat, B. T., Bui, M. Q., & Ngo, V. M. (2024). Automating attendance management in human resources: A design science approach using computer vision and facial recognition. *International Journal of Information Management Data Insights*, 4(2), 100253.
- [4] Thanoon, M. I. (2024). ARTIFICIAL INTELLIGENCE-BASED SMART CLASS ATTENDANCE SYSTEM: AN IOT INFRASTRUCTURE. *International Journal for Quality Research*, 18(1).
- [5] Dechen, K., Pem, K., Dorji, T., Wangdi, T., Powdyel, D. B., & Muramatsu, K. (2024, March). Attendance Management System Using Face Recognition and E-Mail Notification. In *2024 Third International Conference on Intelligent Techniques in Control, Optimization and Signal Processing (INCOS)* (pp. 1-7). IEEE.
- [6] Ali, M., Diwan, A., & Kumar, D. (2024). Attendance System Optimization through Deep Learning Face Recognition. *International Journal of Computing and Digital Systems*, 15(1), 1527-1540.
- [7] Painuly, K., Bisht, Y., Vaidya, H., Kapruwan, A., & Gupta, R. (2024, January). Efficient Real-Time Face Recognition-Based Attendance System with Deep Learning Algorithms. In *2024 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE)* (pp. 1-5). IEEE.
- [8] Oluchukwu, U. E., Okechukwu, O. P., Ezurika, E. O., & Uchefuna, C. I. (2024). Intelligent Face Recognition Based Students' Attendance System. *International Research Journal of Innovations in Engineering and Technology*, 8(4), 8.
- [9] Othman, M. A., Husin, H. S., & Ismail, S. (2024, January). MySIMS: A Hybrid Application of Face Recognition Attendance and Tuition Management System. In *2024 18th International Conference on Ubiquitous Information Management and Communication (IMCOM)* (pp. 1-8). IEEE.
- [10] Gaur, S., Kumar, A., Nirwan, R., Bhowmik, C., & Jain, N. (2024). Biometric Based Attendance System with Machine Learning Integrated Face Modelling and Recognition. In *Advancement of Intelligent Computational Methods and Technologies* (pp. 120-125). CRC Press.

- [11] Ojo, O. S., Oyediran, M. O., Bamgbade, B. J., Adeniyi, A. E., Ebong, G. N., & Ajagbe, S. A. (2023). Development of an improved convolutional neural network for an automated face based university attendance system. *ParadigmPlus*, 4(1), 18-28.
- [12] Jha, P. B., Basnet, A., Pokhrel, B., Pokhrel, B., Thakur, G. K., & Chhetri, S. (2023). An Automated Attendance System Using Facial Detection and Recognition Technology. *Apex Journal of Business and Management*, 1(1), 103-120.
- [13] Saraswat, D., Bhattacharya, P., Shah, T., Satani, R., & Tanwar, S. (2023). Anti-spoofing-enabled contactless attendance monitoring system in the COVID-19 pandemic. *Procedia Computer Science*, 218, 1506-1515.
- [14] Kamil, M. H. M., Zaini, N., Mazalan, L., & Ahamad, A. H. (2023). Online attendance system based on facial recognition with face mask detection. *Multimedia Tools and Applications*, 82(22), 34437-34457.
- [15] Dang, T. V. (2023). Smart attendance system based on improved facial recognition. *Journal of Robotics and Control (JRC)*, 4(1), 46-53.
- [16] Trabelsi, Z., Alnajjar, F., Parambil, M. M. A., Gochoo, M., & Ali, L. (2023). Real-time attention monitoring system for classroom: A deep learning approach for student's behavior recognition. *Big Data and Cognitive Computing*, 7(1), 48.
- [17] Yang, H., & Han, X. (2020). Face recognition attendance system based on real-time video processing. *IEEE Access*, 8, 159143-159150.
- [18] Khan, S., Akram, A., & Usman, N. (2020). Real time automatic attendance system for face recognition using face API and OpenCV. *Wireless Personal Communications*, 113(1), 469-480.
- [19] Dev, S., & Patnaik, T. (2020, September). Student attendance system using face recognition. In *2020 international conference on smart electronics and communication (ICOSEC)* (pp. 90-96). IEEE.
- [20] Bah, S. M., & Ming, F. (2020). An improved face recognition algorithm and its application in attendance management system. *Array*, 5, 100014.
- [21] Seelam, V., kumar Penugonda, A., Kalyan, B. P., Priya, M. B., & Prakash, M. D. (2021). Smart attendance using deep learning and computer vision. *Materials Today: Proceedings*, 46, 4091-4094.
- [22] Pandimurugan, V., Jain, A., & Sinha, Y. (2020, December). IoT based face recognition for smart applications using machine learning. In *2020 3rd International Conference on Intelligent Sustainable Systems (ICISS)* (pp. 1263-1266). IEEE.